Register Number: 192425119

Question 4: Traffic Density by Accident Severity

Research Question

Does traffic density differ significantly among low, medium, and high accident severity groups?

Hypotheses

- H0: μ_Low = μ_Medium = μ_High
- H1: at least one group mean differs
- Significance level α = 0.05

Formula

$F = MS_B / MS_W$

$MS_B = SS_B / (k - 1)$

$MS_W = SS_W / (N - k)$

$SS_B = \Sigma\, n_i\, (\bar{x}_i - \bar{x})^2$

$SS_W = \Sigma\, (n_i - 1)\, s_i^2$

Data Used (read from reference figure)

Severity	n	Mean (≈ median)	SD (≈ IQR/1.35)
Low	38	≈ 40.0	≈ 7.8
Medium	39	≈ 51.5	≈ 10.7
High	23	≈ 66.7	≈ 8.7

Calculation

- Group sizes come from the severity totals in Figure 2 (Low 38, Medium 39, High 23).
- Group means are approximated by the boxplot medians, and SDs by IQR/1.35.
- SS_B ≈ 10,263; SS_W ≈ 8,268; MS_B ≈ 5,132; MS_W ≈ 85.2; F ≈ 60.2 with df = (2, 97).

Result

Quantity	Value
F statistic	60.2
df (between, within)	2, 97
p-value	< 0.001

Conclusion

The p-value is far below 0.05, so we reject H0. Traffic density differs significantly among the Low, Medium, and High severity groups.

Interpretation Notes

- The boxplots show a clear upward shift in median density: about 40 for Low, 51.5 for Medium, and 67 for High.
- Because this ANOVA uses summary values read from the figure, run aov() with the raw data in R and follow up with Tukey's HSD to see which pairs differ.

Graph

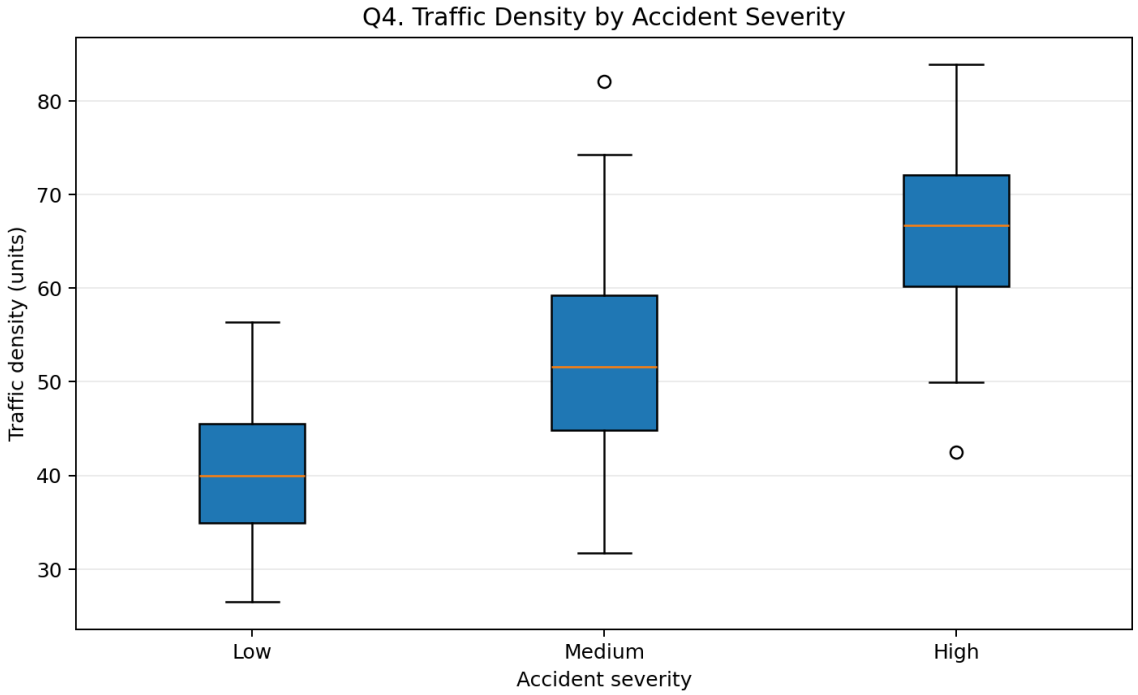


Figure 4. Reference graph for Question 4.

Note: the counts and values in this document were read from the reference figure. Re-run the test on the original project dataset in R (e.g. chisq.test, t.test, aov, cor.test) to confirm the final numbers before submission.